Sheet 14

Collapse noise and relative states

1. A labelled rigid pointer has $N = 10^{23}$, $\lambda_{\mathrm{GRW}} = 10^{-16}\,\mathrm{s}^{-1}$, $r_C = 10^{-7}\,\mathrm{m}$ and separation $8r_C$. Compute the hit and damping rates and first-hit mean waiting time. For the support model take $\ell = 0.02r_C$ and bound the Gaussian tail using $\mathrm{erfc}(a) \le e^{-a^2}/(a\sqrt{\pi})$ for $a > 0$ (given). Apply the lecture's bound at $\epsilon = 0.01$. What changes if separation is only $0.1r_C$? Distinguish the rate calculation from branch selection.
2. **GRW kinetic-energy injection.** The one-hit random kick has zero mean and variance $\hbar^2/(2r_C^2)$ per Cartesian component. For a free particle of finite initial kinetic energy, compute its one-hit mean-energy change and the 1D and 3D rates at hit rate λ_{GRW}. Sum for N equal labelled masses, then estimate the 3D power using the parameters above, $m = 1.67 \cdot 10^{-27}\,\mathrm{kg}$ and $\hbar = 1.055 \cdot 10^{-34}\,\mathrm{J\,s}$. Why is this not directly a temperature rise or a current experimental exclusion claim?
3. In the capstone use input $\rho = \mathbb{1}_2/2$ and purification $\Omega = (|0_S 0_R\rangle + |1_S 1_R\rangle)/\sqrt{2}$. Write the four branch vectors using E_j, the $\pi/3$ rotation and M_k of L10. Compute their weights. For each projected branch, what operator remains if only the two probes are traced out, and what further trace gives the system relative state?